Chapter 6: Covariance and the Multivariate Gaussian Distribution

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CSAI2121 Probability and Statistics
Macao Polytechnic University

Learning Objectives

By the end of this lecture, you should be able to:

- Define **covariance** and the **correlation coefficient**, and interpret their sign and magnitude
- Compute covariance and correlation from a joint distribution
- Explain why **zero correlation does not imply independence** (in general)
- Construct and interpret a **covariance matrix** for a random vector
- Read the **multivariate Gaussian** formula and interpret $\boldsymbol{\mu}$, Σ, and the quadratic form
- Connect covariance/correlation and the multivariate Gaussian to AI/ML applications: feature correlation, anomaly detection, Gaussian models

Random Vectors and the Need for Covariance

Motivation: Beyond Yes/No Independence

- Last week: given a joint distribution, we could only ask **"Are X and Y independent?"** — a yes/no question.
- In Chapter 5, the CNN/Transformer example was *almost* independent (0.24 vs. 0.25) — but we had no way to measure **how close**.
- Many AI questions need more: *How strongly* are two features related? *In what direction*?
- Today: tools to measure the **strength** and **direction** of a relationship — covariance, correlation, and their generalization to many variables.

Recall: Single-Variable Mean and Variance

- Mean: $\mu_X = E[X]$ — the "center" of the distribution.
- Variance: $\mathrm{Var}(X) = E[(X - \mu_X)^2]$ — how spread out X is *around its own mean*.

Today's Extension

We now ask: how does X vary **together** with another variable Y? This requires a new quantity involving *both* variables at once.

Random Vectors: Basic Setup

- A **random vector** collects several random variables into one object:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_d \end{bmatrix}$$

- AI examples: pixel intensities of an image, multiple vital signs of a patient, several features in a dataset row.
- Today we build the theory mainly for $d = 2$ (i.e., X and Y), then generalize to general d.

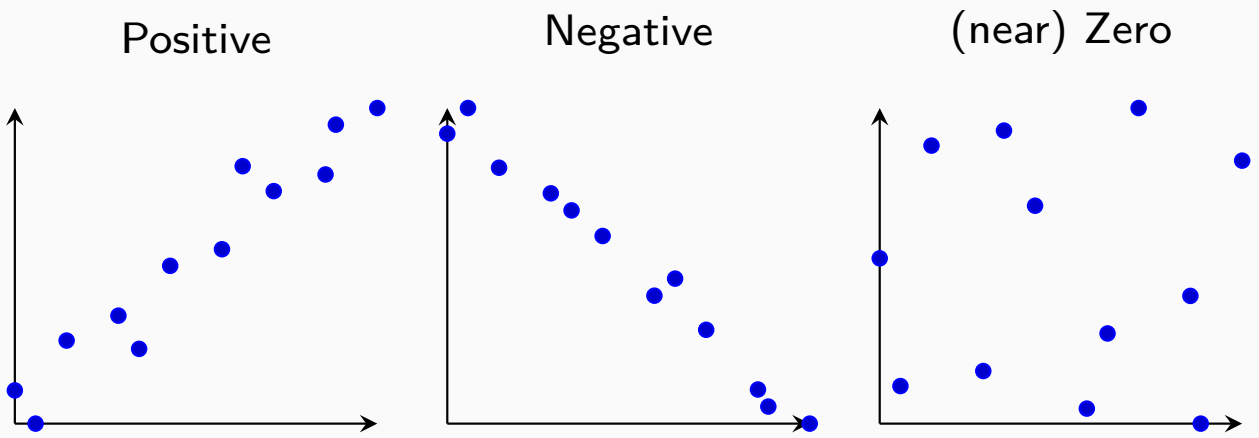
Mean Vector: Definition

Definition

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_d] \end{bmatrix}$$

- No new idea here — just take the expectation of **each component separately**.
- The mean vector tells us *where the distribution is centered*, but nothing about how the variables relate to each other.

Visualizing Relationships: Three Scatter Plots



- Left: X and Y tend to increase together. Middle: X increases as Y decreases. Right: no clear pattern.
- **Question:** how do we turn this visual impression into a *number*?

Covariance and Correlation

Covariance: Definition

Definition

$$\mathrm{Cov}(X, Y) = E\big[(X - \mu_X)(Y - \mu_Y)\big]$$

- For each outcome: is X *above* its mean at the same time Y is *above* its mean (or both below)? Then the product $(X - \mu_X)(Y - \mu_Y)$ is positive.
- If one is above and the other below, the product is negative.
- Covariance is the **average** of this product over all outcomes.

Covariance: Computational Formula

Equivalent (often easier) formula

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

- Same idea as the shortcut formula $\mathrm{Var}(X) = E[X^2] - \mu_X^2$ from Chapter 3–4.
- In fact, $\mathrm{Var}(X) = \mathrm{Cov}(X, X)$ — variance is just a variable's covariance with itself.

Interpreting the Sign of Covariance

- $\mathrm{Cov}(X, Y) > 0$: X and Y tend to move in the **same** direction (Slide 7, left panel).
- $\mathrm{Cov}(X, Y) < 0$: X and Y tend to move in **opposite** directions (Slide 7, middle panel).
- $\mathrm{Cov}(X, Y) = 0$: no *linear* tendency either way (Slide 7, right panel).

Caution

Covariance only measures **linear** tendency. We will see a striking counterexample shortly (Slide 16).

Worked Example: Covariance from a Joint PMF

Let $X, Y \in \{1, 2\}$ with joint PMF:

	$Y = 1$	$Y = 2$	Row sum
$X = 1$	0.40	0.10	0.50
$X = 2$	0.10	0.40	0.50
Col sum	0.50	0.50	1.00

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$$E[XY] = (1)(1)(0.4)+(1)(2)(0.1)+(2)(1)(0.1)+(2)(2)(0.4) = 2.4$$

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$$E[XY] = (1)(1)(0.4)+(1)(2)(0.1)+(2)(1)(0.1)+(2)(2)(0.4) = 2.4$$

$$\mathrm{Cov}(X, Y) = 2.4 - (1.5)(1.5) = 2.4 - 2.25 = 0.15$$

The Problem with Covariance: Units

- Suppose X is height measured in **cm**, and we convert to **m**: $X' = 0.01X$.
- Then: $\mathrm{Cov}(X', Y) = 0.01\ \mathrm{Cov}(X, Y)$ — the number changes even though the *relationship* did not.

General scaling rule

$$\mathrm{Cov}(aX, bY) = ab\ \mathrm{Cov}(X, Y)$$

Key Point

Covariance's magnitude depends on the **units** of X and Y, so it cannot tell us how *strong* a relationship is by itself. We need a unit-free version.

Correlation Coefficient: Definition

Definition

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \, \sigma_Y}$$

- Dividing by $\sigma_X \sigma_Y$ cancels out the units — ρ is always a pure number.
- Property: $-1 \leq \rho_{X,Y} \leq 1$.
- $\rho = 1$: perfect positive linear relationship; $\rho = -1$: perfect negative; $\rho = 0$: no linear relationship.

Worked Example: Correlation (Continuing Slide 11)

$$E[X^2] = 1(0.5) + 4(0.5) = 2.5 \;\Rightarrow\; \mathrm{Var}(X) = 2.5 - 1.5^2 = 0.25$$

Worked Example: Correlation (Continuing Slide 11)

$$E[X^2] = 1(0.5) + 4(0.5) = 2.5 \;\Rightarrow\; \mathrm{Var}(X) = 2.5 - 1.5^2 = 0.25$$

By symmetry, $\mathrm{Var}(Y) = 0.25$, so $\sigma_X = \sigma_Y = 0.5$.

Worked Example: Correlation (Continuing Slide 11)

$$E[X^2] = 1(0.5) + 4(0.5) = 2.5 \;\Rightarrow\; \mathrm{Var}(X) = 2.5 - 1.5^2 = 0.25$$

By symmetry, $\mathrm{Var}(Y) = 0.25$, so $\sigma_X = \sigma_Y = 0.5$.

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{0.15}{0.5 \times 0.5} = \frac{0.15}{0.25} = 0.6$$

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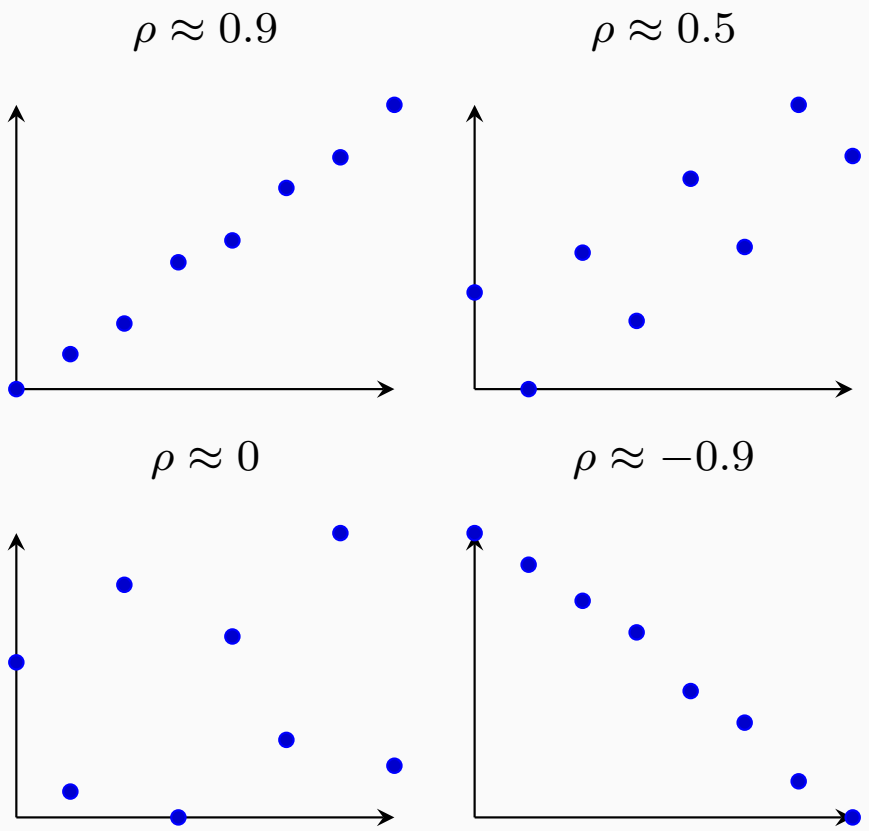
By symmetry, $\mathrm{Var}(Y) = 0.25$, so $\sigma_X = \sigma_Y = 0.5$.

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Interpretation

$\rho = 0.6$: a moderately strong **positive** linear relationship between X and Y.

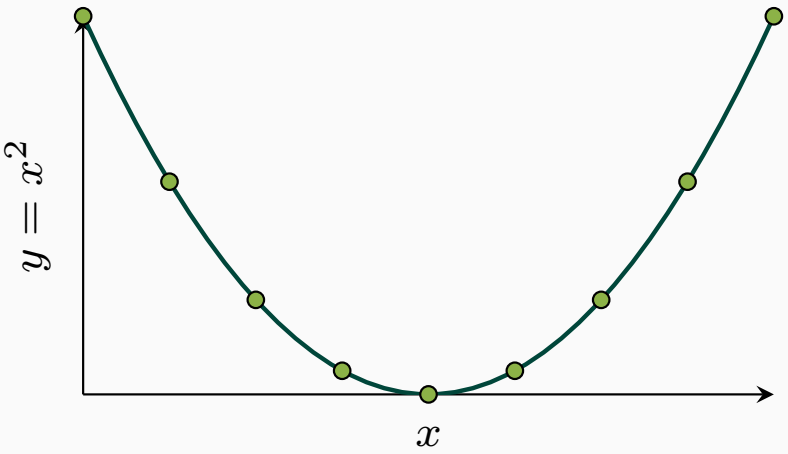
Visualizing Different Correlation Strengths



- As $|\rho| \to 1$, points align more tightly along a line; as $\rho \to 0$, the pattern disappears.

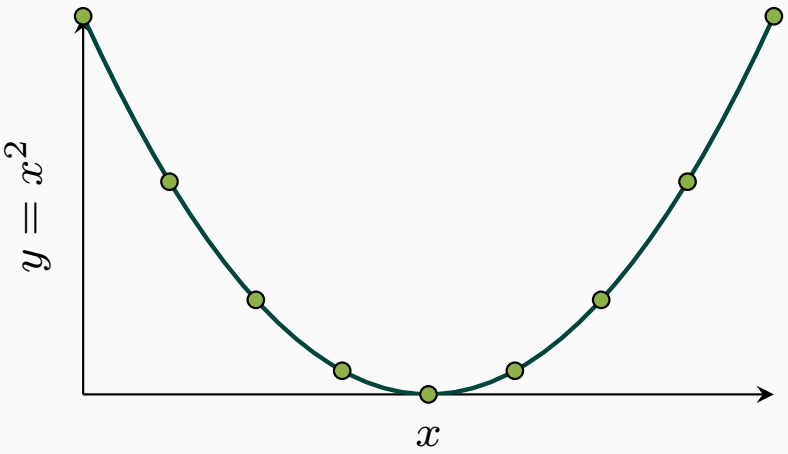
Common Misconception: Zero Correlation vs. Independence

Let X be symmetric around 0 (e.g., uniform on $[-2, 2]$), and $Y = X^2$.



Common Misconception: Zero Correlation vs. Independence

Let X be symmetric around 0 (e.g., uniform on $[-2, 2]$), and $Y = X^2$.



- By symmetry, $E[XY] = E[X^3] = 0$ and $E[X] = 0$, so $\mathrm{Cov}(X, Y) = 0 - 0 = 0$.
- Yet Y is **completely determined** by X — clearly **not independent**!

Key Point

Covariance/correlation only detects **linear** relationships. $\mathrm{Cov} = 0$

AI Application: Feature Correlation in Datasets

- Before training a model, data scientists often compute a **correlation matrix** across all features.
- Highly correlated features (e.g., $|\rho| > 0.9$) may be redundant — a candidate for removal.



- Darker cell = stronger (positive) correlation. Income & Spending: $\rho = 0.8$ — strongly related.

The Covariance Matrix

Motivation: More Than Two Variables

- With d variables $X_1, \ldots, X_d$, there are $\binom{d}{2}$ **pairs**, each with its own covariance.
- Keeping track of every $\mathrm{Cov}(X_i, X_j)$ separately becomes unwieldy as d grows.
- Natural solution: organize all variances and covariances into a single **matrix**.

Covariance Matrix: Definition

Definition

$$\Sigma = \mathrm{Cov}(\mathbf{X}) = E\big[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T\big]$$

- Entry (i, j) of Σ is $\mathrm{Cov}(X_i, X_j)$.
- When $i = j$: $\mathrm{Cov}(X_i, X_i) = \mathrm{Var}(X_i)$.
- This is the natural multivariate generalization of a single variance number.

Covariance Matrix: Structure for $d = 2$

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) \end{bmatrix}$$

- **Diagonal** entries: each variable's own variance.
- **Off-diagonal** entries: pairwise covariances.
- Since $\mathrm{Cov}(X_1, X_2) = \mathrm{Cov}(X_2, X_1)$, the matrix is **symmetric**: $\Sigma = \Sigma^T$.

Properties of the Covariance Matrix

- **Symmetric**: $\Sigma_{ij} = \Sigma_{ji}$ for all i, j.
- **Diagonal = variances**: always ≥ 0.
- ✓ (Extension, not examined) **Positive semi-definite**: for any vector $\mathbf{a}$, $\mathbf{a}^T \Sigma \mathbf{a} \geq 0$.

Intuition for positive semi-definiteness

$\mathbf{a}^T \Sigma \mathbf{a} = \mathrm{Var}(\mathbf{a}^T \mathbf{X})$ — the variance of *any* linear combination of the X_i's. Variance can never be negative, so this must hold.

Worked Example: A 2×2 Covariance Matrix

Using $\mathrm{Var}(X) = \mathrm{Var}(Y) = 0.25$ and $\mathrm{Cov}(X, Y) = 0.15$ from Slides 11 and 14:

$$\Sigma = \begin{bmatrix} 0.25 & 0.15 \\ 0.15 & 0.25 \end{bmatrix}$$

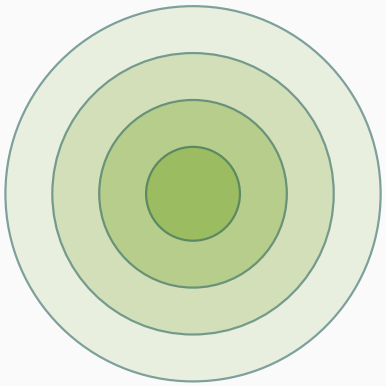
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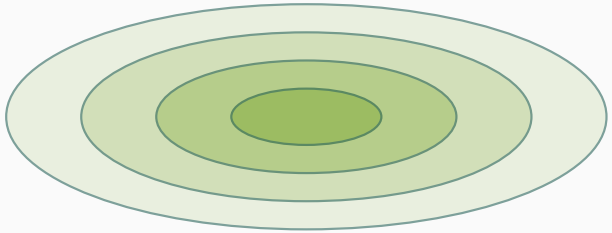
$$\Sigma = \begin{bmatrix} 0.25 & 0.15 \\ 0.15 & 0.25 \end{bmatrix}$$

- Equal diagonal entries: X and Y have the **same spread**.
- Positive off-diagonal entry: they move **together**.
- As we will soon see, this matrix corresponds to a **tilted ellipse** when plotted as a density contour.

Covariance Matrix and the Shape of Data



$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



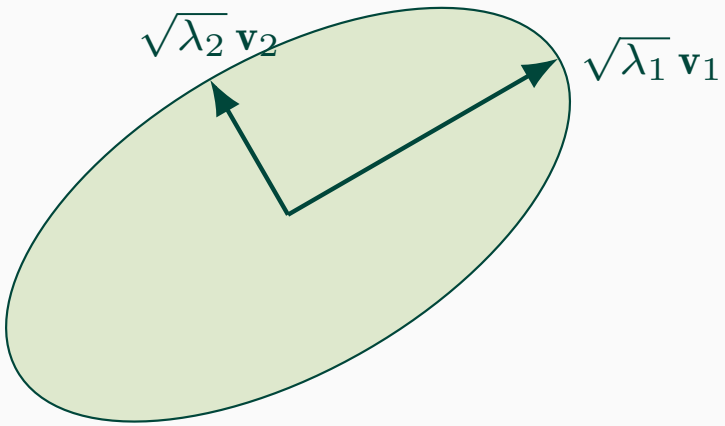
$$\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 0.5 \end{bmatrix}$$



$$\Sigma = \begin{bmatrix} 1 & 0.7 \\ 0.7 & 1 \end{bmatrix}$$

- **Left:** equal variances, zero covariance $\rightarrow$ circular spread.
- **Middle:** unequal variances, zero covariance $\rightarrow$ axis-aligned ellipse.
- **Right:** nonzero covariance $\rightarrow$ **tilted** ellipse.

Instructor Demo: Principal Axes (Preview of PCA)



- Every covariance matrix has **principal axes** (eigenvectors $\mathbf{v}_1, \mathbf{v}_2$), with lengths given by $\sqrt{\lambda_1}, \sqrt{\lambda_2}$ (square roots of eigenvalues).
- These axes point along the directions of **maximum** and **minimum** spread of the data.

✓ This idea is the foundation of **Principal Component Analysis (PCA)**, covered in later AI courses. Not required for this course's assessments.

The Multivariate Gaussian Distribution

From 1D to Multivariate Gaussian

Recall the 1D Gaussian PDF (Chapter 4):

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

From 1D to Multivariate Gaussian

Recall the 1D Gaussian PDF (Chapter 4):

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- Note: $\dfrac{(x-\mu)^2}{\sigma^2} = (x-\mu)(\sigma^2)^{-1}(x-\mu)$ — a "squared distance", scaled by the variance.
- **Idea for generalization:** replace x, μ with vectors $\mathbf{x}, \boldsymbol{\mu}$, and σ^2 with the covariance matrix Σ.

Multivariate Gaussian: The Formula

Definition

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

- This is the natural vector/matrix generalization of the 1D formula above.
- We will **not** derive the normalizing constant $(2\pi)^{d/2}|\Sigma|^{1/2}$ — it simply ensures the total probability integrates to 1.

Interpreting Each Term in the Formula

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}\underbrace{(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}_{\text{Mahalanobis distance}^2}\right)$$

- $\boldsymbol{\mu}$: the **center** of the distribution.
- Σ: controls the **shape** (spread and tilt) of the distribution.
- $(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})$: a "distance" from $\mathbf{x}$ to $\boldsymbol{\mu}$ that accounts for the shape of Σ.
- $|\Sigma|^{1/2}$: part of the normalizing constant.

Special Case: Diagonal Covariance Matrix

If $\Sigma = \begin{bmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{bmatrix}$, the formula **factorizes**:

$$p(x_1, x_2) = p(x_1) \cdot p(x_2)$$

where each $p(x_i)$ is an ordinary 1D Gaussian.

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Resolving Slide 16's Misconception — A Special Case

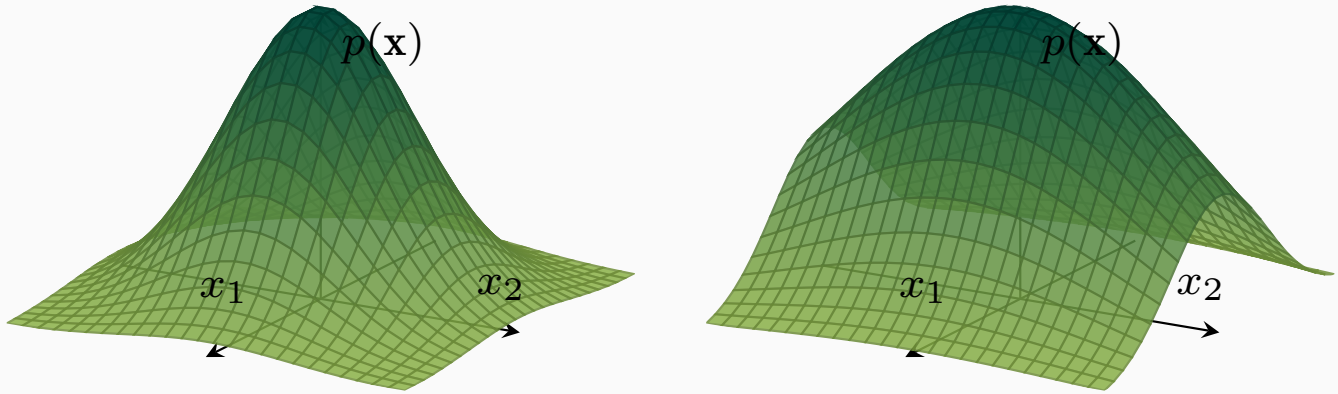
For **jointly Gaussian** random variables (but *not* in general!):

$$\mathrm{Cov}(X, Y) = 0 \implies X, Y \text{ independent.}$$

This is a special property of the Gaussian family, not a general rule.

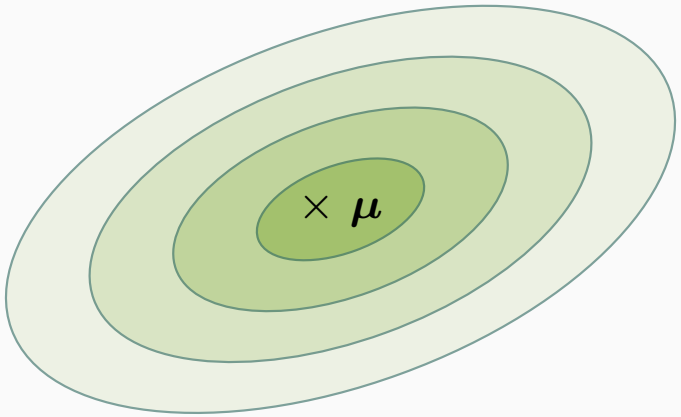
Visualizing the Bivariate Gaussian as a Surface

$\Sigma = I$ (circular) $\Sigma = \mathrm{diag}(1, 4)$ (elongated)



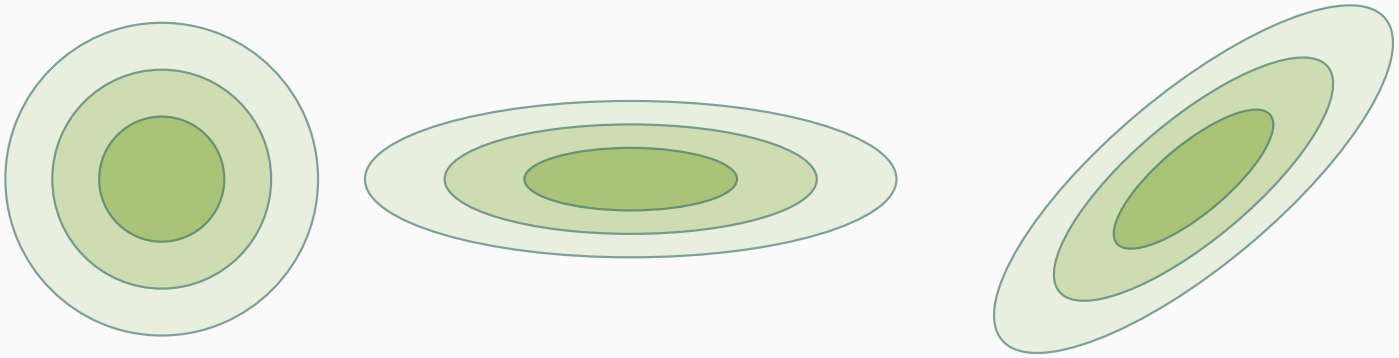
- Left: equal variances $\rightarrow$ symmetric "bell-shaped hill". Right: larger variance along x_2 $\rightarrow$ stretched hill.

Contour Plot of the Bivariate Gaussian



- Each ellipse is a set of points with the **same density value**: $(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) = c$ for some constant c.
- This is exactly the same ellipse family we saw for the covariance matrix in Slide 23 — Σ **determines the shape of the Gaussian's contours**.

How μ and Σ Control the Shape



shift $\boldsymbol{\mu}$ change variances add covariance (tilt)

- Changing $\boldsymbol{\mu}$: the whole "hill" **translates**, shape unchanged.
- Changing diagonal entries of Σ: the ellipse **stretches** along the axes.
- Adding off-diagonal covariance: the ellipse **tilts**.

Extension: The Mahalanobis Distance

Definition

$$D_M(\mathbf{x})^2 = (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

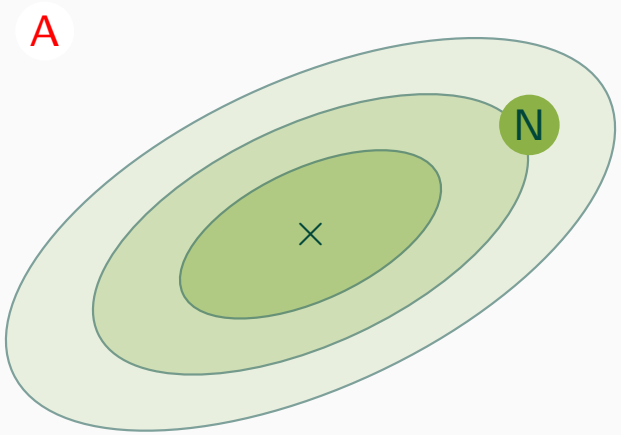
- Unlike ordinary Euclidean distance, Mahalanobis distance accounts for **how spread out** the data is in each direction.
- A point that is "close" in raw coordinates may be **far** in Mahalanobis distance if it lies against the natural spread of the data (and vice versa).

✓ Full derivation and theoretical justification are **not required** for this course.

AI Application: Anomaly Detection using Mahalanobis Distance

- Fit $\boldsymbol{\mu}, \Sigma$ from "normal" sensor data (e.g., temperature X_1, humidity X_2).
- For a new reading $\mathbf{x}$, flag as **anomalous** if:

$$D_M(\mathbf{x})^2 = (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) > \tau \quad (\tau = \text{threshold})$$



- Point "N" lies within the typical ellipse $\rightarrow$ normal. Point "A" lies far outside $\rightarrow$ flagged as anomaly, even though its raw coordinates are not extreme in either dimension alone.

AI Application: Gaussian Assumptions in Machine Learning

- **Gaussian Naive Bayes**: assumes each feature, given the class, follows a (univariate) Gaussian.
- **Gaussian Mixture Models (GMM)**: model complex data as a *mixture* of several multivariate Gaussians (introduced in later AI courses).
- **Kalman filters** (robotics, tracking): assume sensor noise and system state follow multivariate Gaussians.
- **Anomaly/outlier detection**: as just shown, via Mahalanobis distance.

Key Point

The multivariate Gaussian is one of the most widely used building blocks in probabilistic AI models — largely because $\boldsymbol{\mu}$ and Σ give a compact, interpretable description of a whole distribution.

Summary and Looking Ahead

Summary: Key Formulas from Today

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = E[XY] - \mu_X\mu_Y$$

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}, \qquad -1 \le \rho_{X,Y} \le 1$$

$$\Sigma = E\big[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^T\big] \quad \text{(symmetric, } \Sigma_{ii} = \mathrm{Var}(X_i)\text{)}$$

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T\Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

Common Mistakes

- Comparing covariance values computed on **different units** and concluding one relationship is "stronger"
- Concluding **independence** just because $\mathrm{Cov}(X, Y) = 0$ (only valid, in general, for jointly Gaussian variables)
- Forgetting that the covariance matrix must be **symmetric**
- Treating Mahalanobis distance as if it were ordinary Euclidean distance
- Assuming correlation captures **nonlinear** relationships (it does not)

Connections Ahead

- Covariance/correlation (this week) $\rightarrow$ foundation for **linear regression** (Week 9) and **PCA**
- Covariance matrix $\rightarrow$ core object in **Gaussian Mixture Models**, **Kalman filters**, **multivariate statistics**
- Multivariate Gaussian $\rightarrow$ noise model behind **OLS regression** and many probabilistic ML methods
- Mahalanobis distance $\rightarrow$ outlier/anomaly detection in real AI systems

Class Exercise

Problem

1. A joint PMF table for $X, Y \in \{0, 1\}$ is given in the handout. Compute $\mathrm{Cov}(X, Y)$ and $\rho_{X,Y}$.
2. Given $\Sigma = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, is this a valid covariance matrix? Sketch the approximate shape of its density contours (circle / axis-aligned ellipse / tilted ellipse), and state the tilt direction.

Work in pairs (15 minutes), then we discuss as a class.

Class Exercise: Solution to Problem 2

$$\Sigma = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

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Step 1: Is Σ valid?

- Symmetric? ✓ ($\Sigma_{12} = \Sigma_{21} = -1$)
- Diagonal entries ≥ 0? ✓ (both are 2)
- Eigenvalues: $\lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$ — both > 0 ✓

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$\Rightarrow$ **Valid** (in fact positive *definite*) covariance matrix.

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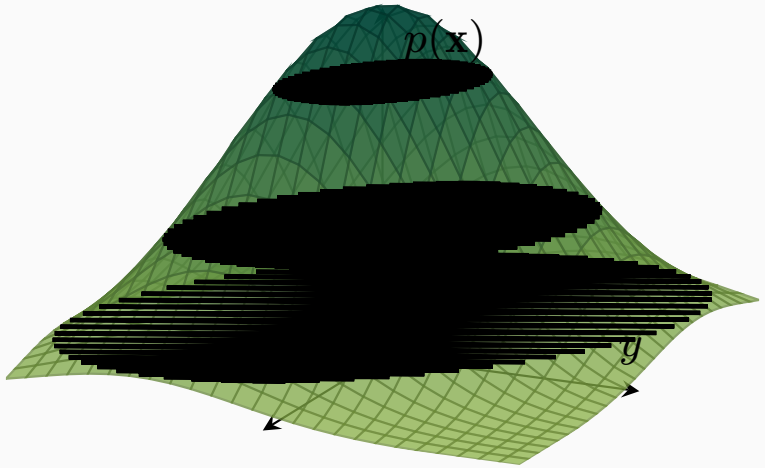
$\Rightarrow$ **Valid** (in fact positive *definite*) covariance matrix. **Step 2: Shape and tilt.**

$$\rho_{X,Y} = \frac{-1}{\sqrt{2}\sqrt{2}} = -0.5 \quad \text{(negative correlation)}$$

Eigenvector for $\lambda = 3$ (major axis): $(1, -1)/\sqrt{2} \Rightarrow$ ellipse tilted along the line $y = -x$.

Class Exercise: Solution to Problem 2

$$\Sigma = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$



Conclusion

The density contours form a **tilted ellipse**, elongated along $y = -x$ — consistent with X and Y tending to move in **opposite** directions ($\rho = -0.5$).

Preview of Next Week: Law of Large Numbers and CLT

- Today: how to describe the shape and relationship structure of a **single** joint distribution.
- Next week: what happens when we draw **many i.i.d. samples** from a distribution and look at their **average**?
- We will meet the **Law of Large Numbers** (the average settles near the true mean) and the **Central Limit Theorem** (the average becomes approximately Gaussian).
- Essential groundwork for confidence intervals and hypothesis testing later in the course.